

JORGE CHAVEZ

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Education

University of Illinois at Urbana-Champaign

Aug 2019 – May 2025

Master of Science in Electrical and Computer Engineering – GPA: 3.83/4.00

Graduated May 2025

– Concentration in AI Safety & Machine Learning Verification

Bachelor of Science in Computer Engineering – GPA: 3.83/4.00

Graduated May 2023

Projects

[\$\alpha, \beta\$ -CROWN, Neural Network Verification](#) | PyTorch Based Verifier

Jan 2024 – Present

- Engineered a branch-and-bound optimization framework that **cut subproblems by 96%** and **accelerated verification by up to 85%**, driving the team to **1st place in VNN-COMP 2024 & 2025 (#1 on all 21 benchmarks)**, including complex modern architectures like the **Vision Transformer (ViT)** robustness benchmark. All experiments deployed on standardized AWS A10G GPU instances to ensure reproducibility and cloud-scalable evaluation.
- Implemented a PGD-based soundness check integrated into the project's CI/CD pipeline **to automate validation of new contributions**, with support for local runs to streamline development.

Bounding Neural Network Implicit Surfaces | PyTorch

Sept 2024 – Present

- Developed Guaranteed **Inner-and-Outer Meshes (GIOM)**, a framework combining neural network verification with neural implicit surfaces to accelerate geometric queries in computer graphics.
- Achieved **3x faster rendering** with **5 dB PSNR improvement**, **5x faster** collision detection, and **10x lower** constructive solid geometry (CSG) reconstruction errors compared to prior methods with minimal computational overhead.
- Integrated the open-source, state-of-the-art neural verification tool (α, β -CROWN) to produce certified bounding meshes, **enabling sound and efficient ray intersection, collision detection, and constructive solid geometry operations**.
- Co-authored research paper** (*submission to ICLR 2026*), contributing theoretical guarantees and practical benchmarks for high-fidelity neural surface rendering.

Work Experience

Motorola Solutions

Jan 2022 – May 2022

Fullstack Developer

Champaign, IL

- Engineered and deployed a full-stack internal hiring dashboard **using Angular and Azure DevOps to streamline the recruiting process** of students at UIUC's Research Park.
- Designed and implemented a responsive recruiter interface that enabled real-time retrieval of resumes, efficient planning of campus recruiting events, and tracking of attendee data.
- Collaborated within a team of four, working closely with backend engineers to define and shape API requirements, ensuring seamless data flow and system integration.

Motorola Solutions

May 2021 – Aug 2021

Backend Developer

Champaign, IL

- Optimized database querying and ETL processes** (MySQL and Elastic Stack) to standardize service metrics used across production monitoring dashboards.
- Prepared **high-efficiency backend data views** (MySQL) to report real-time service uptime, ticketing status, and latency for the Network Operations Center (NOC).
- Coordinated with global engineering and operations teams (North America, Europe, Asia) to implement unified monitoring dashboards and establish cross-functional data standards.

Leadership Experience

Illini Space Society | Controls Programmer and Project Lead

Sept 2024 – May 2025

- Supervised a 3-member team** to design and deploy feedback controllers for autonomous systems using PyTorch (NNs) and PyBullet (simulation), enabling robust and effective trajectory stabilization.
- Applied both ML-based and classical approaches to achieve stable autonomous behavior: **engineered neural-network controllers with Lyapunov stability guarantees** (verified via α, β -CROWN) for inverted pendulum/cart-pole systems, and implemented LQR to achieve stable quadrotor control at arbitrary equilibrium points.
- Presented technical workshops on state-space modeling, feedback control, and Kalman filtering, **mentoring 15 peers and strengthening the organization's expertise in control theory**.

Publications

- [Clip-and-Verify: Linear Constraint-Driven Domain Clipping for Accelerating Neural Network Verification](#)

Neural Information Processing Systems (NeurIPS), 2025 - accepted, equal contribution

- [Guaranteed Bounding Meshes Extraction from Neural Implicit Surfaces via Neural Network Verification](#)

International Conference on Learning Representations (ICLR), 2026 - under review, equal contribution

- [A Linear Constraint Driven Approach to Efficiently Enhancing Branch and Bound in Neural Network Verification](#)

Master's Thesis, UIUC, 2025

Programming Skills: C++, Python, MySQL, Matlab, Verilog, CUDA, PyTorch, Angular, OpenCV, Pandas, ROS